Algebra 7 — Solving quadratic equations · Quadratic inequalities

Calculator allowed. Marks in brackets. ★ = harder question. Quadratic formula: $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

1 Solve by factorising. [5]

a $3x^2 - 48 = 0$ **b** $2x^2 + 5x - 3 = 0$ **c** $x(3x - 7) = 6$

2 **a** Complete the square for $x^2 - 6x + 4$. **b** Hence solve $x^2 - 6x + 4 = 0$, giving your answers in surd form. [4]

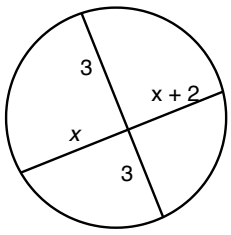
3 Use the quadratic formula to solve, giving answers in surd form. [4]

a $x^2 + 3x - 2 = 0$ **b** $3x^2 - 5x + 1 = 0$

4 Solve each quadratic inequality and show the solution set on a number line. [4]

a $3x^2 > 27$ **b** $x^2 - 2x - 8 \leq 0$

5 Two chords of a circle cross as shown. **a** Use the intersecting chords theorem to form an equation in x. **b** Solve it to find x, correct to 3 significant figures. [4]



6 A rectangular poster is 4 cm longer than it is wide. Its area is 60 cm². Find its width. [4]

Total: 25 marks

Check: bring everything to one side (= 0) before factorising · completing the square: $(x + \frac{b}{2})^2 - (\frac{b}{2})^2 + c$ · formula: write a, b, c with signs, b^2 is positive, $-4ac$ positive if a or c is negative · inequality: sketch, roots, "below" = between, "above" = outside · reject negative lengths.